Module 2: Essentials of Cloud Computing and Characteristics

Cloud Computing

- Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.
- You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.

https://azure.microsoft.com/en-au/overview/what-is-cloud-computing/#benefits

The NIST Definition of Cloud Computing

- According to National Institute of Standards and Technology: Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- This cloud model is composed of five essential characteristics, three service models, and four deployment models.

Essential Characteristics

- **On-Demand Self-Service:**
  Cloud computing allows users to automatically provision resources like computing power, storage, and networking without human intervention from the service provider.
- **Broad Network Access:**
  Cloud services are accessible over the network (internet) using a variety of devices such as smartphones, laptops, and tablets. This ensures flexibility and availability.
- **Resource Pooling:**
  Cloud providers use multi-tenant models to pool resources and distribute them dynamically across different users. This helps in efficient resource management and cost reduction.

Essential Characteristics

- **Rapid Elasticity:**

  Cloud systems can quickly scale resources up or down according to demand. This elasticity ensures that users only pay for what they use, adjusting dynamically as needed.

- **Measured Service:**

  Cloud computing resources are metered and billed based on usage. This pay-as-you-go model ensures that users only pay for the services they use, often based on usage metrics like compute time or storage space.

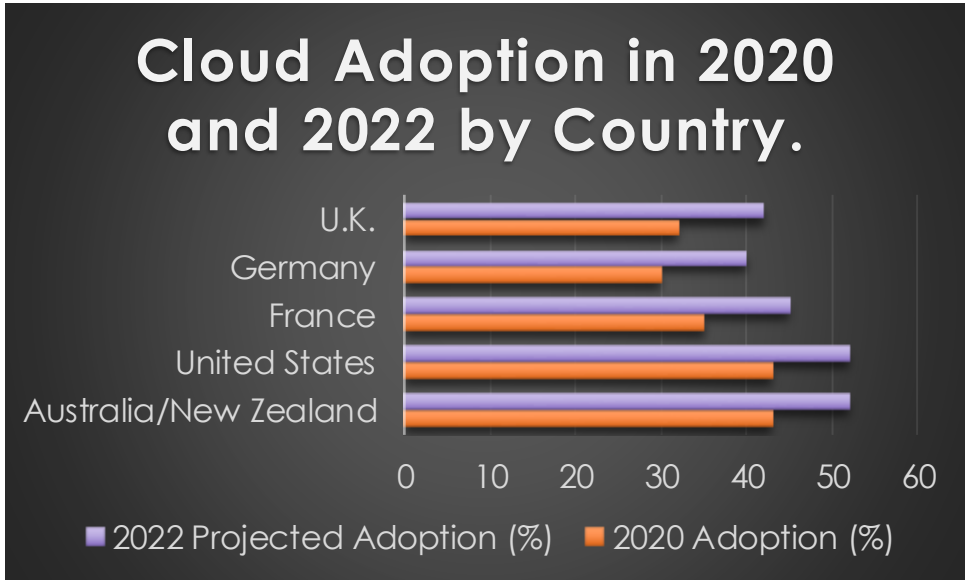
Learning Activity 1:
Race to Fill In the Five Essentials of Cloud

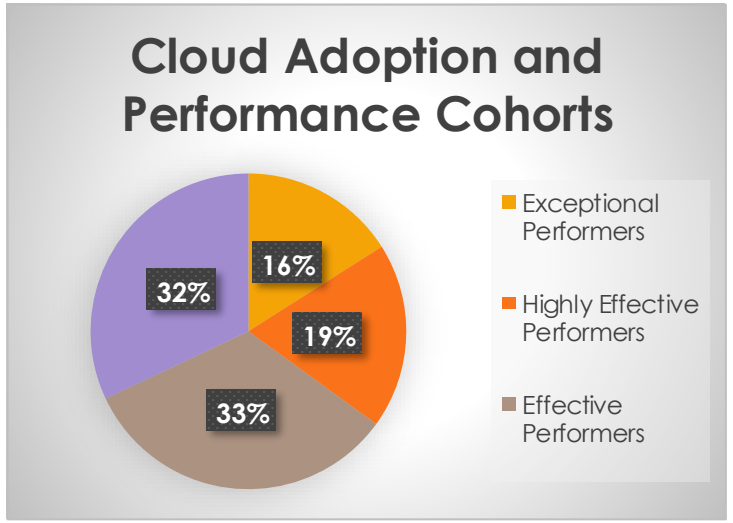
- Post your reflection in the Module 2 discussion forum.
- Access Module 2: Learning Activity 1 - Discussion Forum

Learning Activity 2:

📌 Cloud Adoption in Australia & New Zealand: Insights & Challenges

- 🎯 Key Takeaways from the Article:
- - 🌍 Australia & NZ are cloud leaders but still face adoption challenges.
- - 🏛️ Government policies encourage cloud growth.
- - 🔒 Security, compliance & cost remain top concerns.
- - 🤖 AI & automation are shaping cloud usage.





Difference between

Parameter	Amazon Web Services (AWS) – Elastic Compute Cloud (EC2)	Microsoft Azure – Azure AI & Machine Learning
Service Type	Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)
Primary Function	Provides scalable virtual computing instances for running applications and workloads.	Offers machine learning and AI tools for building and deploying intelligent applications.
Target Customers	Businesses needing compute power for applications, websites, and big data processing.	Businesses looking to implement AI, automation, and data-driven decision-making in their operations.
Scalability	High scalability for computing power with instant provisioning of virtual servers (instances).	Scalable resources for AI model training, inference, and deployment across cloud infrastructure.
Core Benefit	Flexible and cost-efficient cloud compute capacity with a pay-as-you-go model.	Simplifies AI model development and deployment, even for those with limited expertise in AI.
Cost Model	Pay-as-you-go pricing based on the instance type and usage.	Consumption-based pricing based on resource usage for training, inferencing, and storage.
Popular Use Cases	Web hosting, app development, data processing, and high-performance computing.	Predictive analytics, data-driven decision-making, and automation via AI-powered apps.
Integration with Other Services	Can be integrated with AWS services like S3 for storage, RDS for databases, and Lambda for serverless functions.	Integrates with Azure services like Power BI for data analysis, Azure Databricks for data engineering, and Cognitive Services for pre-built AI models.
Global Availability	Available in multiple AWS regions worldwide with low latency and high redundancy.	Available globally, with data centers in multiple regions supporting hybrid cloud solutions.
Customer Experience	Self-service management via AWS Management Console, providing deep customization and flexibility.	Easy-to-use interfaces and pre-built models to accelerate AI application development, available through Azure Studio and APIs.

Learning Activity 3:

Key Cloud Computing Services Offered by the Cloud Service Provider

- Cloud service providers are generally those companies that:

  - Establish and manage public clouds
  - Manage private clouds
  - Offer on-demand cloud computing services such as Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service(SaaS)

- For example, Amazon Web Services (AWS), Microsoft Azure and Google Cloud Platform (GCP) are some of the well-known service providers. To get an appropriate understanding of cloud services, please watch this video about Current Major Cloud ProvidersLinks to an external site..

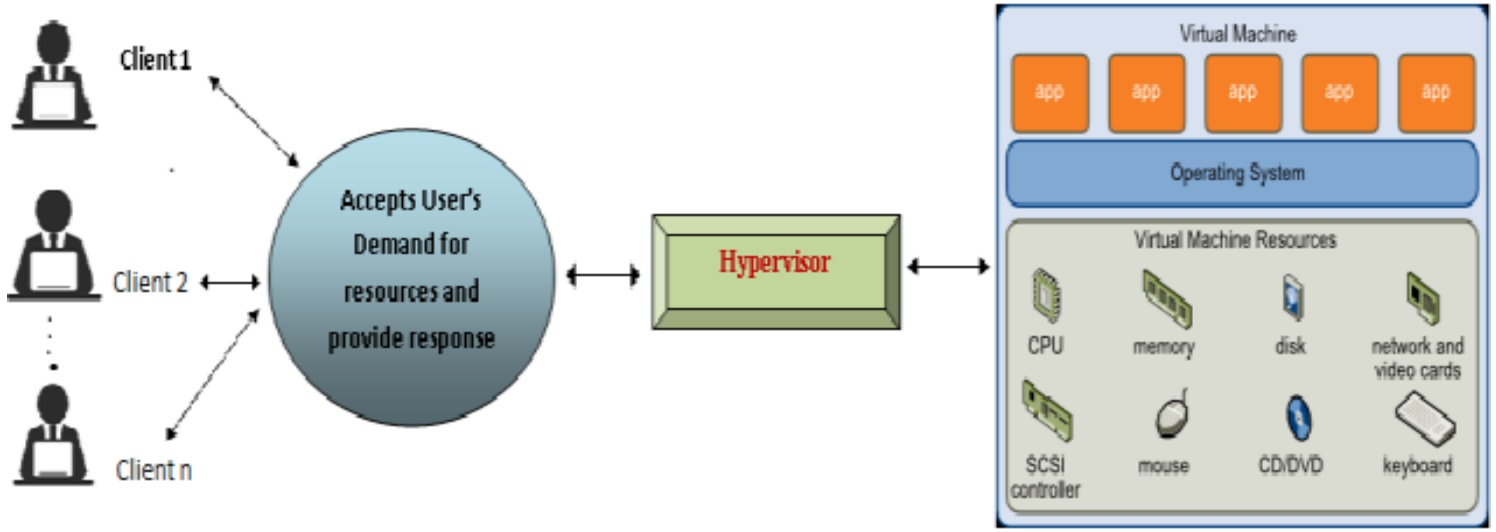
- Now that you are more familiar with cloud service providers and services, select any two cloud service providers from AWS, Microsoft Azure, IBM Cloud or GCP and discuss at least one key service offered by them. Why do you think this key service unlocks so much value for the customer?

- Post your reflection in the Module 2 discussion forum.

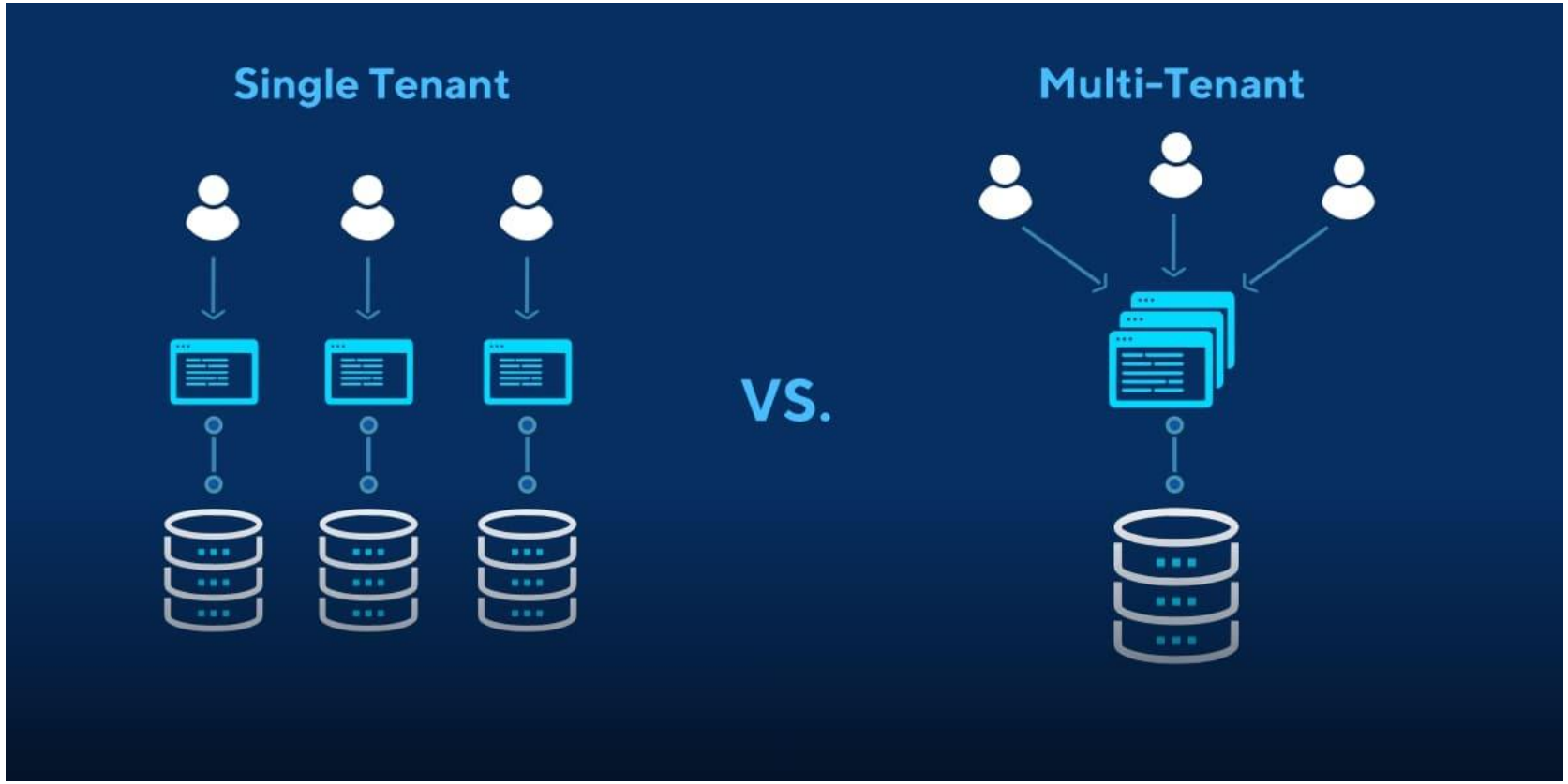
- Access Module 2: Learning Activity 3 - Discussion Forum

Essential Characteristics

Resource pooling.

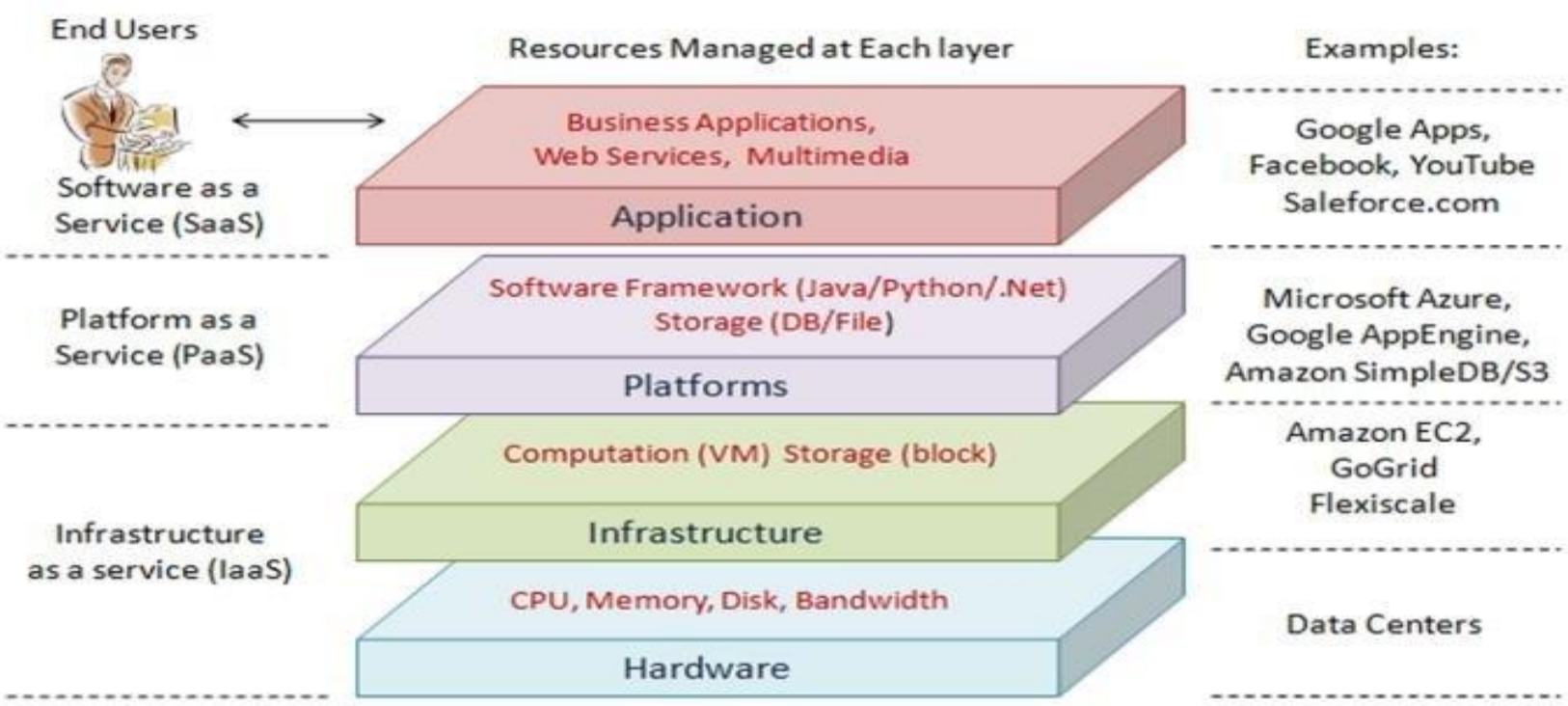


Rashid, A., & Chaturvedi, A.K. (2017). A Study on Resource Pooling, Allocation and Virtualization Tools used for Cloud Computing. *International Journal of Computer Applications*, *168*, 7-11.



https://www.eescorporation.com/multi-tenancy-issues-in-cloud-computing/

Layered cloud computing architecture



https://www.researchgate.net/figure/Layered-cloud-computing-architecture-9_fig4_265793878

Cloud service models

Table 1.2 *Cloud service models*

Service model	Capability offered to the user	Controllability by users
Software as a service (SaaS)	Use of applications that run on the cloud.	Limited application configuration settings, but no control over underlying cloud infrastructure – network, servers, operating systems, storage, or individual application capabilities.
Platform as a service (PaaS)	Deployment of applications on the cloud infrastructure; may use supported programming languages, libraries, services, and tools.	The user has control of deployed applications and their environment settings, but no control of cloud infrastructure – network, servers, operating systems, or storage.
Infrastructure as a service (IaaS)	Provisioning of processing, storage, networks, etc.; may deploy and run operating systems, applications, etc.	The user has control of operating systems, storage, and deployed applications running on virtualized resources assigned to the user, but no control over underlying cloud infrastructure.

Service Models:

Software as a Service (SaaS). The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure[2]. The applications are accessible from various client devices through either a thin client interface, such as a web browser (e.g., web-based email), or a program interface. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

Software as a Service | SaaS

- It is a software distribution model in which services are hosted by a cloud service provider.
- following services provided by SaaS providers

Popular SaaS Providers

https://www.javatpoint.com/software-as-a-service

Popular SaaS Providers

Provider	Services
Salseforce.com	On-demand CRM solutions
Microsoft Office 365	Online office suite
Google Apps	Gmail, Google Calendar, Docs, and sites
NetSuite	ERP, accounting, order management, CRM, Professionals Services Automation (PSA), and e-commerce applications.
GoToMeeting	Online meeting and video-conferencing software
Constant Contact	E-mail marketing, online survey, and event marketing
Oracle CRM	CRM applications
Workday, Inc	Human capital management, payroll, and financial management.

https://www.javatpoint.com/software-as-a-service

Benefits of SaaS

- The main benefit of SaaS is that it offloads all infrastructure and application management to the SaaS vendor. All the user has to do is create an account, pay the fee and start using the application. The vendor handles everything else, from maintaining the server hardware and software to managing user access and security, storing and managing data, implementing upgrades and patches and more.
- Minimal risk: Many SaaS products offer a free trial period, or low monthly fees that let customers try the software to see if it will meet their needs, with little or no financial risk.
- Anytime/anywhere productivity: Users can work with SaaS apps on any device with a browser and an internet connection.
- Easy scalability: Adding users is as simple as registering and paying for new seats; customers can purchase more data storage for a nominal charge.

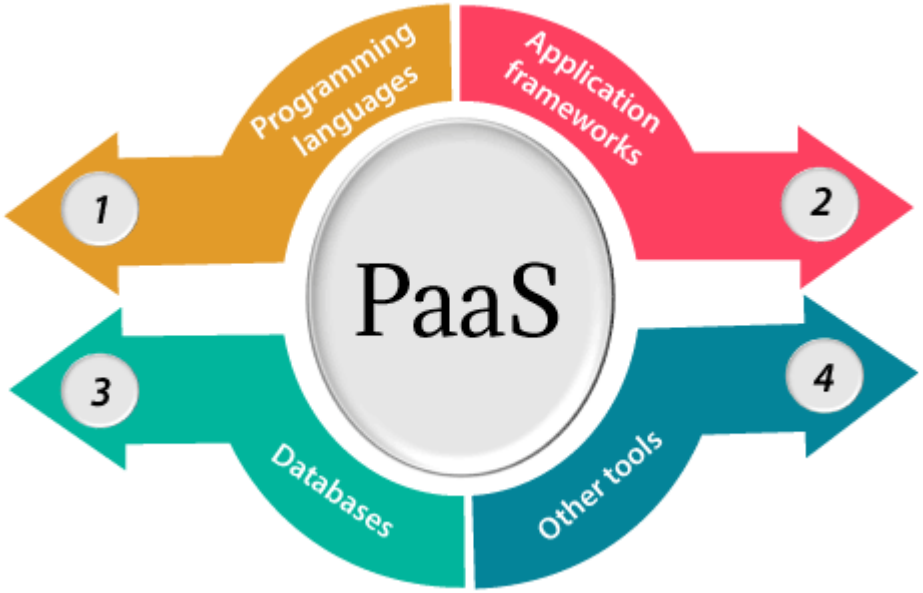
https://www.ibm.com/cloud/learn/iaas-paas-saas

Service Models:

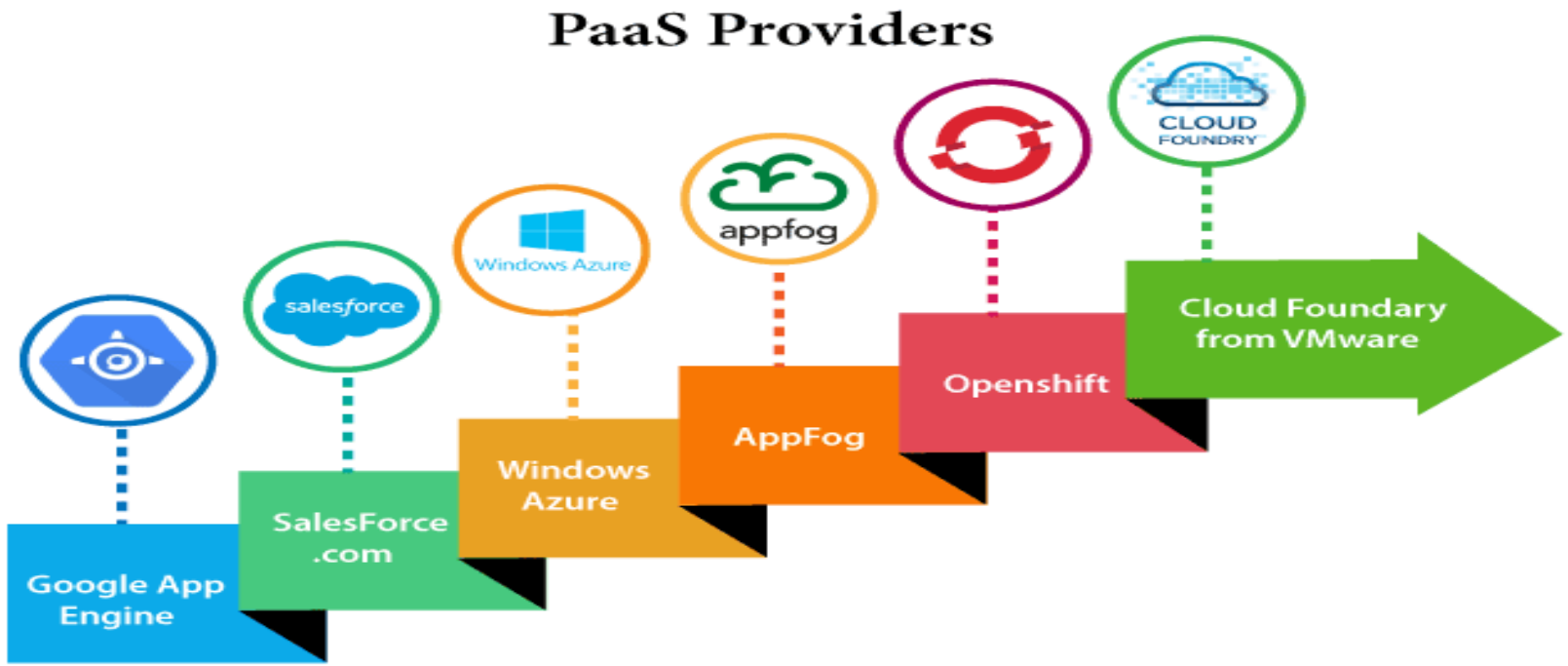
- **Platform as a Service (PaaS).** The capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages, libraries, services, and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has control over the deployed applications and possibly configuration settings for the application-hosting environment.

Platform as a Service | PaaS

- Platform as a Service (PaaS) provides a runtime environment. It allows programmers to easily create, test, run, and deploy web applications.



Popular PaaS Providers



Popular PaaS Providers

Providers	Services
Google App Engine (GAE)	App Identity, URL Fetch, Cloud storage client library, Logservice
Salesforce.com	Faster implementation, Rapid scalability, CRM Services, Sales cloud, Mobile connectivity, Chatter.
Windows Azure	Compute, security, IoT, Data Storage.
AppFog	Justcloud.com, SkyDrive, GoogleDocs
Openshift	RedHat, Microsoft Azure.
Cloud Foundry from VMware	Data, Messaging, and other services.

Benefits of PaaS

- The primary benefit of PaaS is that it allows customers to build, test, deploy run, update and scale applications more quickly and cost-effectively than they could if they had to build out and manage their own on-premises platform.
- Faster time to market: PaaS enables development teams to spin-up development, testing and production environments in minutes, vs. weeks or months.
- Low- to no-risk testing and adoption of new technologies: PaaS platforms typically include access to a wide range of the latest resources up and down the application stack. This allows companies to test new operating systems, languages, and other tools without having to make substantial investments in them, or in the infrastructure required to run them.

https://www.ibm.com/cloud/learn/iaas-paas-saas

Benefits of SaaS

- Simplified collaboration: As a cloud-based service, PaaS provides a shared software development environment, giving development and operations teams access to all the tools they need, from anywhere with an Internet connection.
- A more scalable approach: With PaaS, organizations can purchase additional capacity for building, testing, staging and running applications whenever they need it.
- Less to manage: PaaS offloads infrastructure management, patches, updates and other administrative tasks to the cloud service provider.

https://www.ibm.com/cloud/learn/iaas-paas-saas

Service Models:

Infrastructure as a Service (IaaS). The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., host firewalls).

Infrastructure as a Service | IaaS

- It allows customers to outsource their IT infrastructures such as servers, networking, processing, storage, virtual machines, and other resources. Customers access these resources on the Internet using a pay-as-per use model.

Iaas Providers

IaaS Vendor	Iaas Solution	Details
Amazon Web Services	Elastic, Elastic Compute Cloud (EC2) MapReduce, Route 53, Virtual Private Cloud, etc.	The cloud computing platform pioneer, Amazon offers auto scaling, cloud monitoring, and load balancing features as part of its portfolio.

Benefits of IaaS

- Compared to traditional IT, IaaS gives customers more flexibility build out computing resources as needed, and to scale them up or down in response to spikes or slow-downs in traffic.
- IaaS lets customers avoid the up-front expense and overhead of purchasing and maintaining its own on-premises data center.
- It also eliminates the constant trade-off between the waste of purchasing excess on-premises capacity to accommodate spikes, versus the poor performance or outages that can result from not having enough capacity for unanticipated traffic bursts or growth.

https://www.ibm.com/cloud/learn/iaas-paas-saas

Benefits of IaaS

- Higher availability: With IaaS a company can create redundant servers easily, and even create them in other geographies to ensure availability during local power outages or physical disasters.
- Lower latency, improved performance: Because IaaS providers typically operate data centers in multiple geographies, IaaS customers can locate apps and services closer to users to minimize latency and maximize performance.
- Improved responsiveness. Customers can provision resources in a matter of minutes, test new ideas quickly and quickly roll out new ideas to more users.

https://www.ibm.com/cloud/learn/iaas-paas-saas

Benefits of IaaS

- Comprehensive security. With a high-level of security on-site, at data centers, and via encryption, organizations can often take advantage of more advanced security and protection they could provide if they hosted the cloud infrastructure in-house.
- Faster access to best-of-breed technology. Cloud providers compete with each other by providing the latest technologies to their users, IaaS customers can take advantage of these technologies much earlier (and at far less cost) than they can implement them on premises.

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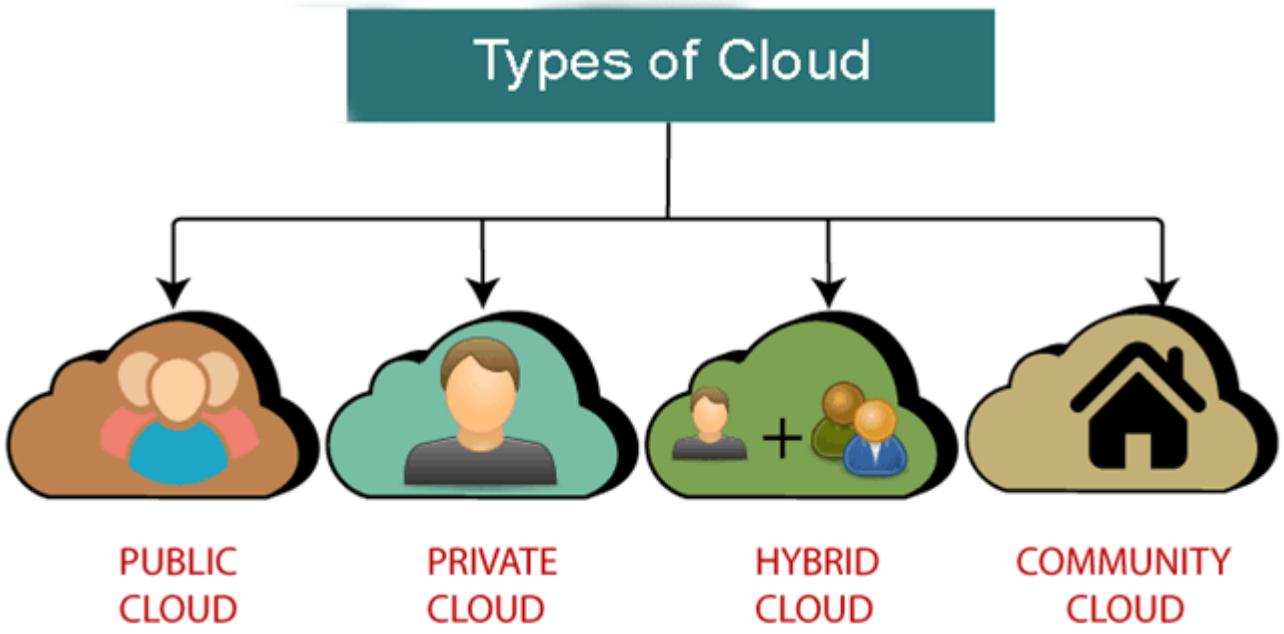
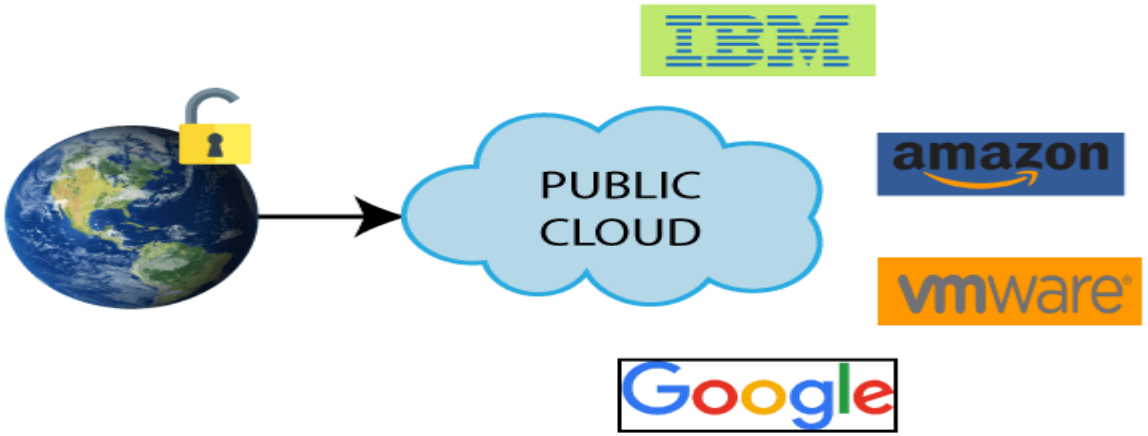
Deployment Models



https://www.javatpoint.com/types-of-cloud

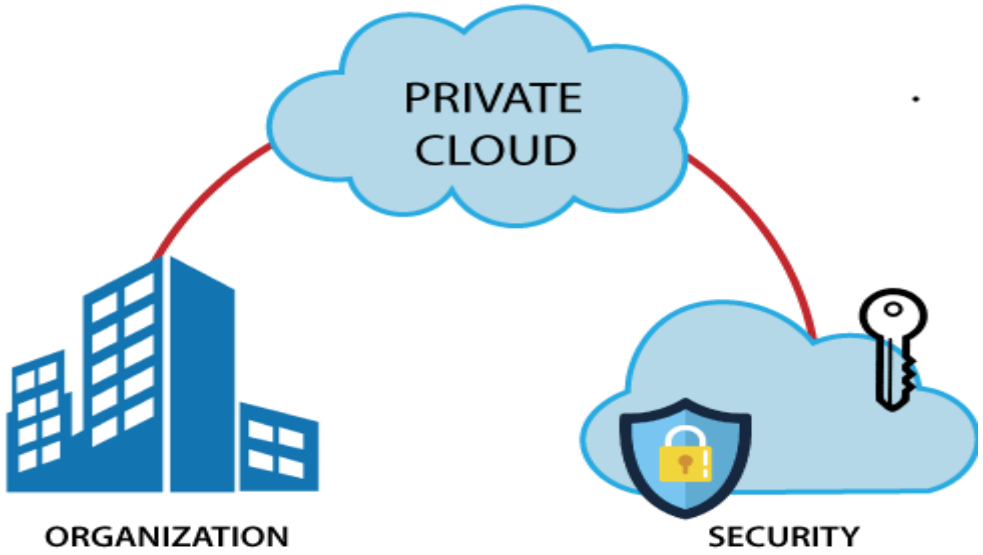
Public Cloud

- Public cloud is open to all to store and access information via the Internet using the pay-per-usage method.
- Example: Amazon elastic compute cloud (EC2), IBM SmartCloud Enterprise, Microsoft, Google App Engine, Windows Azure Services Platform.



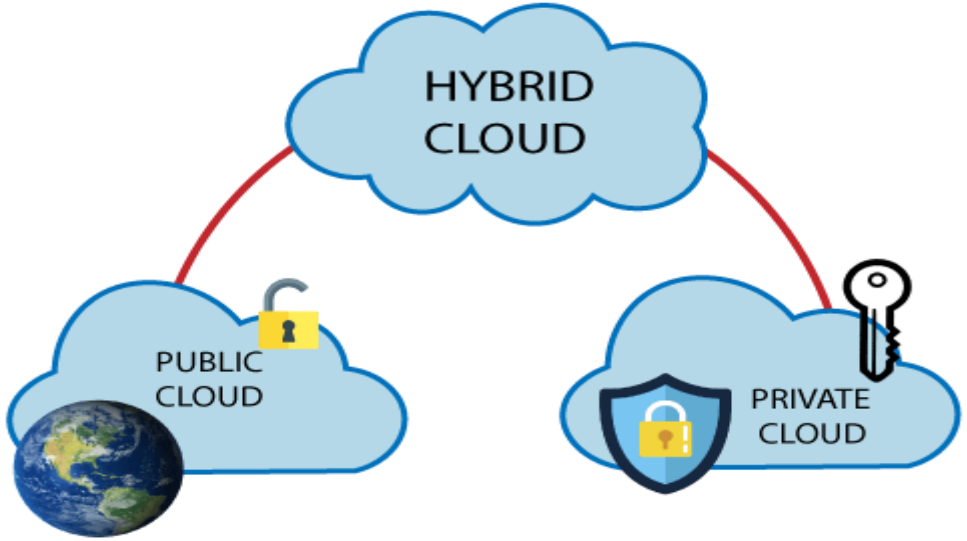
Private Cloud

- Private cloud is also known as an internal cloud or corporate cloud. It is used by organizations to build and manage their own data centers internally or by the third party



Hybrid Cloud

- Hybrid Cloud = Public Cloud + Private Cloud
- Example: Google Application Suite (Gmail, Google Apps, and Google Drive), Office 365 (MS Office on the Web and One Drive), Amazon Web Services.

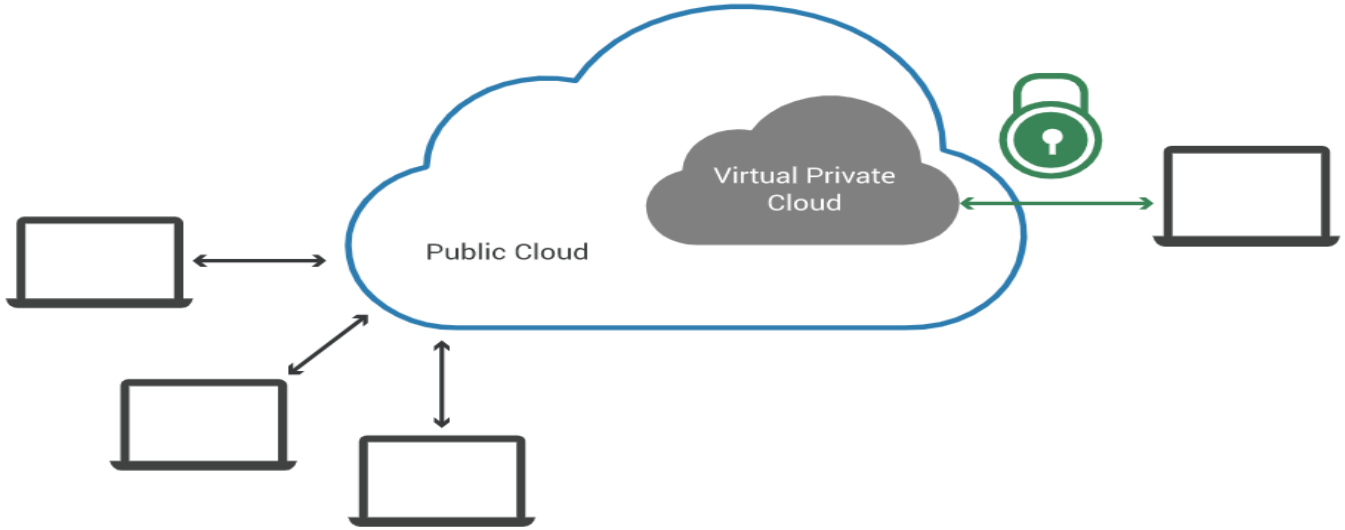


Community Cloud

- Community cloud allows systems and services to be accessible by a group of several organizations to share the information between the organization and a specific community.
- It is owned, managed, and operated by one or more organizations in the community, a third party, or a combination of them. Example: Health Care community cloud

virtual private cloud (VPC)?

- A virtual private cloud (VPC) is a secure, isolated private cloud hosted within a public cloud. VPC customers can run code, store data, host websites, and do anything else they could do in an ordinary private cloud, but the private cloud is hosted remotely by a public cloud provider



Cloud Support Services

- To facilitate transition to the cloud, a cloud ecosystem is emerging that aims to offer a spectrum of new cloud support services that augment, complement, or assist the popular SaaS, IaaS, and PaaS offerings.
- Examples of such cloud support services are data storage as a service (DSaaS), analytics as service (AaaS), desktop as a service (DAAS), security as a service (SecaaS), identity and access management as a service (IAMaaS), and monitoring as a service (MaaS).

Advantages of cloud computing

Access to resources

The greatest advantage of cloud computing is the access it provides to the processing power of multiple remote computers. This enables customers to take advantage of greater computation speed and larger storage capacity than most organisations can provide on their premises and at a fraction of the cost.

Mobility

Customers can access the services from almost any location in the world because the services are web-based (and because of the advent of mobile devices). This can enable employees to access important business tools while they are on the move. For example, the employee can fill in a Tempora online timesheet whilst on a train, providing the rest of the business with access to that data in real time.

Advantages of cloud computing

Easily scalable

Both the monthly subscription and 'pay as you use' charging models make it easy for the amount of service being provided to be increased or decreased. Should a customer want to increase the number of 'seats' included in its subscription to Tempora or the amount of megabytes of storage space rented from AWS, this can be done easily. The supplier simply provides access to additional users or increases the storage space available in exchange for higher monthly payments by the customer. The scalability of the cloud computing model makes it especially attractive to growing organisations with varying levels of demand for computer resources (e.g. where an organisation's website receives higher volumes of visitors at certain times of year).

Data security and storage capacity

Data security is of particular importance as lapses in procedure can cause severe financial and reputational damage. For the majority of organisations, the data security and data storage capacity offered by data centres is far superior to that which can be afforded in-house. This is because they specialise in the secure storage of data.

Advantages of cloud computing

Cost savings

The ‘pay as you use’ system is of particular benefit to an organisation with peaks and troughs in its demand for computing resources. It is cheaper than paying for exclusive use of enough resources to meet peak demand when it is not required, as is the case where all computation is carried out by an organisation in-house.

Additionally, cloud services reduce the need for an organisation to maintain in-house expertise in their own technological infrastructure, which reduces IT costs.

Finally, cloud computing services do not represent a capital expenditure, so customers lose less if they switch suppliers.

Advantages of cloud computing

Maintenance and support

The supplier will usually offer ongoing support services. However, remote hosting of the services makes the process of maintaining and supporting the services less intrusive for the customer. The supplier can handle backups, updates and upgrades automatically and remotely without visiting a customer's site. This will generally mean that maintenance and support can be carried out more quickly. In addition, customers are able to piggy-back on their suppliers' upgrades in computing resources and are not locked into using infrastructure purchased at great cost 10 years previously.

Free trials

Some suppliers offer the opportunity to trial their product for a period without charge. This is made easier by the supplier's ability to terminate access at the end of the period and provides them with the opportunity to 'hook' the customer. This business model is sometimes referred to as a 'freemium'.

Disadvantages of cloud computing

Internet reliability

Clearly where IT services are provided over the internet, lack of internet access or slow connections will hinder access to those services. Where those services are business-critical this can be a major problem. However, as internet access improves, this should be a diminishing concern. Also, it should be remembered that there is no guarantee of uninterrupted service even with locally hosted software applications or data storage, which can be rendered inoperable by defects or bugs.

Dependence on the supplier

With cloud computing the customer is dependent on the supplier for day-to-day access to the IT services rather than just for support and maintenance. If the supplier is in financial trouble, is reliant on an unstable subcontractor or is involved in litigation, its ability to provide the services may be affected. These issues could leave the customer without access to business-critical systems.

ADDITIONAL SERVICE MODELS

- **Database as a Service:**
  - DbaaS provides a database platform organizations can use to store their data.
  - Many PaaS providers also provide database services.
- **Desktop as a Service**
  - DaaS provides users with a virtual desktop that can be used to perform desktop computing.

ANY
QUESTIONS